

A native icosian realization of Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$

The 600-cell unit group reproduces a family of cuspidal Hecke systems — parameter-free and out-of-sample

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2026-06-02 (pre-peer-review preprint, v1.0.0-rc1)

Abstract

We give an independent, parameter-free implementation of the standard definite Brandt method for cuspidal Hilbert modular forms over $K = \mathbb{Q}(\sqrt{5})$, driven entirely by the geometry of the 600-cell. The binary icosahedral group $2I$ is the reduced-norm-one unit group of a maximal order $\mathcal{O}$ (the *icosian ring*) in the definite quaternion algebra ramified at the two infinite places of K ; its lattice is E_8 . For a split prime $\mathfrak{p} \subset \mathcal{O}_K$ we build the Brandt module of $\mathcal{O}$ at level $\mathfrak{p}$ purely from the action of $A_5 = 2I/\{\pm 1\}$ on $\mathbb{P}^1(\mathcal{O}_K/\mathfrak{p})$, with no input from elliptic curves, modular symbols, or L -function tables. The resulting Hecke operators are integral and self-adjoint, and we verify against independently point-counted data that this construction reproduces *a family* of forms, not a single example: (i) the complete prime-level cuspidal-dimension sequence over K for norm < 100 (7/7 agreements with Demb  le, including the two non-obvious cases — the absent form at norm 59 and the jump to a genus-2 form at norm 61); (ii) the full Hecke eigenvalue sets of *two* rational eigenforms (levels 31 and 41), matched out-of-sample against brute-force point counts of the associated elliptic curves at every prime tested, with no fitting; (iii) the genus-2 form at level 61, whose irrational eigenvalues in $\mathbb{Z}[\varphi]$ (real multiplication by K) the geometry reconstructs, matching the published values exactly at the inert primes. We close with an explicitly heuristic discussion of how this finite, self-adjoint, coefficient-side structure relates to the Riemann Hypothesis, emphasizing the gap rather than bridging it: RH is the *global, analytic* positivity equivalent to the Weil and Li criteria, of which a finite Brandt computation supplies nothing. **This paper proves no Millennium problem and claims no progress on one.** Its contribution is a clean, reproducible geometric realization of known arithmetic, and an honest delimitation of the gap that remains.

Scope

This is a *reproduction-and-map* paper. The arithmetic it recovers is due to Demb  le [7] and is an instance of the Jacquet–Langlands correspondence [9, 8]; our contribution is the independent, parameter-free *geometric* route (native icosian arithmetic, no computer-algebra modular package), the explicit no-fit out-of-sample protocol, and the honest closure/resonance map of the RH wall (§8). We claim *validation and exposition*, not new number theory, and *no* progress on RH itself.

1 Introduction

The 600-cell is the regular convex 4-polytope whose 120 vertices are the unit icosians: the binary icosahedral group $2I \subset \mathbb{H}$ of order 120. Viewed as a lattice in $\mathbb{R}^4 \cong \mathbb{H}$, the icosians are a copy of

E_8 over $\mathbb{Z}$, and a copy of the Hurwitz-type maximal order $\mathcal{O}$ in the definite quaternion algebra $B = \left(\frac{-1, -1}{K}\right)$ over $K = \mathbb{Q}(\sqrt{5})$ [5, 13]. This is the smallest arithmetic setting in which the golden ratio $\varphi = (1 + \sqrt{5})/2$ and the icosahedral symmetry that organizes the 600-cell become number theory.

By the Jacquet–Langlands correspondence and the Eichler basis problem [9, 8, 12], the Brandt module of $\mathcal{O}$ at a prime level $\mathfrak{p}$ computes the Hilbert cusp forms of parallel weight 2 and level $\mathfrak{p}$ over K . Demb  le [7] used exactly this to produce the first explicit tables of such forms; the smallest level carrying a cusp form has norm 31, and the associated rational eigenform corresponds to an elliptic curve over K [2].

Our question is narrow and testable: *does the 600-cell geometry, used on its own terms, reproduce this arithmetic?* We answer yes, and quantitatively. The validation is anti-circular by construction: every target we compare against is generated independently — by brute-force point counting of an elliptic curve, or from the published literature — and is used *only in comparison*, never in building the Brandt operators.

Results.

1. **One form, completely (§3).** At level $\mathfrak{p}_{31}$ the geometry gives a 2-dimensional Brandt module, Eichler mass $8/15$, a 1-dimensional cuspidal eigenform, and Hecke eigenvalues matching the point-counted curve 31.1-a1 at all 24 good primes of norm ≤ 200 . Resolving the two primes over each split rational prime, the geometry reproduces the *complete* eigenvalue pair $\{a_q, a_{q'}\}$, not merely one representative.
2. **A family, by dimension (§4).** The geometric orbit count reproduces the entire prime-level cuspidal-dimension sequence over K for norm < 100 : 7/7 agreement with [7], including the absence of a form at norm 59 and the jump to cuspidal dimension 2 at norm 61.
3. **A family, by eigenvalue (§5).** A second rational eigenform, at level $\mathfrak{p}_{41}$, is built by the same procedure and matched out-of-sample (full eigenvalue sets, 9/9 primes) against the point-counted curve $E_{41} : y^2 + \varphi y = x^3 - \varphi x^2$.
4. **The non-rational form (§6).** At level $\mathfrak{p}_{61}$ the geometry yields cuspidal dimension 2 — the genus-2 abelian surface with real multiplication by K of [7, Rem. 4.4]. From the 3×3 Brandt matrix we recover, at every prime, an eigenvalue whose field-theoretic discriminant is $5 \cdot (\text{square})$, i.e. an irrational element of $\mathbb{Z}[\varphi]$; at the inert primes the values match the published ones exactly.
5. **The certificate algebra and the RH map (§7–§8).** The Brandt operators are self-adjoint for the orbit-size measure — a closure certificate in a small algebra we make precise. We use the same algebra to locate RH: the automorphic coefficient side is what the geometry delivers; the analytic positivity that is RH is the lossless self-adjoint engine, which we characterize and do not build.

All computations are pure Python, reproduce in seconds to minutes, and are included with the paper; §9 gives the manifest.

2 The icosian order and its Brandt module

Let $K = \mathbb{Q}(\sqrt{5})$, $\varphi = (1 + \sqrt{5})/2$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$, with $N(a + b\varphi) = a^2 + ab - b^2$ and nontrivial automorphism $\sigma : \varphi \mapsto 1 - \varphi$. Let $B = \left(\frac{-1, -1}{K}\right)$ with standard basis $1, i, j, k$. The *icosian ring* $\mathcal{O}$ is the maximal order whose group of reduced-norm-one units is the binary icosahedral group, $\mathcal{O}^\times = 2I$, $|2I| = 120$; as a $\mathbb{Z}$ -lattice $\mathcal{O}$ is E_8 [5, 13]. B is ramified exactly at the two infinite places of K , hence definite, and (its maximal orders form) a single ideal class.

Throughout we take the level $\mathfrak{p} \subset \mathcal{O}_K$ to be a prime that *splits* in K , so that $N(\mathfrak{p}) = q$ is a rational prime and the residue field is $\mathcal{O}_K/\mathfrak{p} \cong \mathbb{F}_q$. (The levels 31, 41, 61 used below are all split; the inert and ramified cases, with residue field $\mathbb{F}_{q^2}$ or $\mathbb{F}_q$, are not needed.) For odd q fix a local splitting

$$\iota_{\mathfrak{p}} : \mathcal{O} \otimes_{\mathcal{O}_K} (\mathcal{O}_K)_{\mathfrak{p}} \xrightarrow{\sim} M_2(\mathcal{O}_K/\mathfrak{p}) = M_2(\mathbb{F}_q),$$

constructed explicitly by solving $i^2 = j^2 = -1$, $ij = -ji$ over $\mathbb{F}_q$ (possible since -1 is a sum of two squares mod q). The unit group $2I$ acts through $\iota_{\mathfrak{p}}$ on the $q+1$ points of $\mathbb{P}^1(\mathbb{F}_q)$; since $-1 \in 2I$ acts trivially, the action factors through $A_5 = 2I/\{\pm 1\}$. Because $\mathcal{O}$ has class number one, the right-ideal classes of the Eichler order $\mathcal{O}_0(\mathfrak{p})$ identify with $\mathcal{O}^1 \backslash \mathbb{P}^1(\mathcal{O}_K/\mathfrak{p})$, i.e. with the A_5 -orbits on $\mathbb{P}^1(\mathbb{F}_q)$ (the definite/Brandt method; [13, Ch. 41], [12]). Write h for the number of orbits and $\mu = (\mu_1, \dots, \mu_h)$ for their sizes, so $\sum_i \mu_i = q+1$.

Brandt operators. For a totally positive $\varpi \in \mathcal{O}_K$ with $N(\varpi) = N(\mathfrak{q})$, $\mathfrak{q} \nmid \mathfrak{p}$, enumerate the icosians of reduced norm ϖ (a short-vector enumeration in E_8). Their number is $120 r(\varpi)$ where $r(\varpi) = N(\mathfrak{q}) + 1$ is the relevant Brandt / theta coefficient for the icosian order at the primes we use; we verify this count at construction time as a gate. Push each element through $\iota_{\mathfrak{p}}$ to a matrix in $\mathrm{GL}_2(\mathbb{F}_q)$, and let it permute $\mathbb{P}^1(\mathbb{F}_q)$. Accumulating the induced map on orbits gives the Brandt matrix $B(\mathfrak{q}) \in M_h(\mathbb{Q})$,

$$B(\mathfrak{q})_{IJ} = \frac{1}{120 \mu_I} \# \{ (x, \gamma) : x \in \text{orbit } I, N(\gamma) = \varpi, x \cdot \iota_{\mathfrak{p}}(\gamma) \in \text{orbit } J \}. \quad (1)$$

These are the Hecke operators $T_{\mathfrak{q}}$ on the Brandt module. Three intrinsic invariants are checked at construction time, before any arithmetic target is consulted:

- *Eisenstein row sums:* $\sum_J B(\mathfrak{q})_{IJ} = N(\mathfrak{q}) + 1$ (the all-ones vector is the Eisenstein eigenvector);
- *integrality:* $B(\mathfrak{q}) \in M_h(\mathbb{Z})$;
- *self-adjointness:* $\mu_I B(\mathfrak{q})_{IJ} = \mu_J B(\mathfrak{q})_{JI}$, i.e. $B(\mathfrak{q})$ is symmetric for the inner product weighted by the orbit sizes μ .

The cuspidal subspace is the μ -orthogonal complement of the Eisenstein vector; on it the eigenvalues of $B(\mathfrak{q})$ are the Hecke eigenvalues $a_{\mathfrak{q}}$ of the cusp forms of level $\mathfrak{p}$.

Scope

Equation (1) and the bijection “classes = A_5 -orbits” are standard for class number one [12, 13]. The point of the paper is that we instantiate them with nothing but icosian short-vector arithmetic and the A_5 -action, and then test the output against independent data.

3 Level 31: one form, completely

At $\mathfrak{p}_{31} = (5\varphi - 2)$, $N(\mathfrak{p}_{31}) = 31$ (the smallest norm carrying a cusp form [7]), the A_5 -action on $\mathbb{P}^1(\mathbb{F}_{31})$ has two orbits, $\mu = (12, 20)$, so $h = 2$ and the cuspidal dimension is 1. The point stabilizers have orders $60/12 = 5$ and $60/20 = 3$, giving Eichler mass $\frac{1}{5} + \frac{1}{3} = \frac{8}{15} = \frac{1}{60}(31+1)$, in agreement with the mass formula (with $|\zeta_K(-1)|/2 = 1/60$ for K). The cuspidal eigenvector is $(3, -5)$.

Out-of-sample match. Independently of the geometry we form the target by brute-force point counting of the elliptic curve 31.1-a1/ K , $[a_1, \dots, a_6] = [1, \varphi + 1, \varphi, \varphi, 0]$, setting $a_q = N(q) + 1 - \#E(\mathcal{O}_K/q)$. No L -function table is read. The Brandt eigenvalues from (1) agree with a_q at all 24 good primes of norm ≤ 200 .

Verified (computational, reproducible)

Split-prime completeness. For a split rational prime $q = qq'$ the two Hecke eigenvalues $a_q, a_{q'}$ are the values at the totally positive generator ϖ and at its conjugate $\sigma(\varpi)$. The geometry reproduces the *full set* $\{a_q, a_{q'}\}$ — not one representative up to the global Galois involution. For $q = 11, \dots, 89$:

q	11	19	29	41	59	61	71	79	89
$\{a_q, a_{q'}\}$	$\{-4, 4\}$	$\{-4, 4\}$	$\{-2\}$	$\{-6\}$	$\{-4, 12\}$	$\{-2, 6\}$	$\{-8, 0\}$	$\{0, 16\}$	$\{-6, 10\}$

matching the point-counted target exactly (singletons at $q = 29, 41$ occur where $a_q = a_{q'}$).

Atkin–Lehner sign. The curve 31.1-a1 has nonsplit multiplicative reduction at $\mathfrak{p}_{31}$ ($\#E_{\text{ns}}(\mathbb{F}_{31}) = q + 1$, i.e. $a_{\mathfrak{p}_{31}} = -1$). With the standard newform convention $W_{\mathfrak{p}} = -a_{\mathfrak{p}}$ at a prime of multiplicative reduction exactly dividing the level, the Atkin–Lehner (Fricke) eigenvalue is $W_{31} = -a_{\mathfrak{p}_{31}} = +1$, consistent with the Brandt data.

4 A family, by dimension

The orbit count of §2 gives the Brandt dimension h , and hence the cuspidal dimension $h - 1$ at every prime level with no further computation. (The Eisenstein subspace at prime level is 1-dimensional because $K = \mathbb{Q}(\sqrt{5})$ has narrow class number one, so there is a single Eisenstein class, the all-ones vector.) Table 1 compares the geometric output with Dembélé’s table [7] at every split prime level in his displayed range ($31 \leq N \leq 89$); the smaller split prime norms (11, 19, 29) all give cuspidal dimension 0, consistent with 31 being the smallest level carrying a cusp form.

	$N(\mathfrak{p})$	31	41	59	61	71	79	89
geometry (cusp. dim)		1	1	0	2	1	1	1
Dembélé [7]		1	1	0	2	1	1	1
type		ell.	ell.	—	gen. 2	ell.	ell.	ell.

Table 1: Prime-level cuspidal dimensions over $\mathbb{Q}(\sqrt{5})$, geometry vs Dembélé. 7/7. The two informative cases are 59 (no weight- $(2, 2)$ cusp form, only Eisenstein [7, Ex. 3.6]) and 61 (the first form with coefficients outside $\mathbb{Q}$, a genus-2 surface [7, Rem. 4.4]).

The point of Table 1 is that the orbit count is not tuned to one example: with no free parameters it reproduces a non-obvious *sequence*, including a vanishing (59) and a jump to a non-rational form (61). The same computation continues to larger split levels at no extra cost; we do not tabulate those here, as we have not cross-checked them against an independent source.

5 A family, by eigenvalue: level 41

We repeat §3 verbatim at $\mathfrak{p}_{41}$ ($N = 41$, split): the A_5 -action on $\mathbb{P}^1(\mathbb{F}_{41})$ has two orbits, $\mu = (12, 30)$, $h = 2$, cuspidal dimension 1. The cuspidal eigenvalues are integral, self-adjoint, and satisfy the

Ramanujan bound $|a_q| \leq 2\sqrt{N(q)}$ at every prime tested.

For the out-of-sample target we point-count Dembélé's curve $E_{41} : y^2 + \varphi y = x^3 - \varphi x^2$ ([7, Tab. 4], $[a_1, \dots, a_6] = [0, -\varphi, \varphi, 0, 0]$).

Verified (computational, reproducible)

Second form, out-of-sample, no fit. Geometric eigenvalue sets $\{a_q, a_{q'}\}$ vs point-counted E_{41} :

q	11	19	29	31	59	61	71	79	89
geom	$\{-2, 5\}$	$\{-1, 6\}$	$\{2, 9\}$	$\{-10, 4\}$	$\{-3, 4\}$	$\{-8, 6\}$	$\{-12, 9\}$	$\{-11, -4\}$	$\{-8, -1\}$
E_{41}	$\{-2, 5\}$	$\{-1, 6\}$	$\{2, 9\}$	$\{-10, 4\}$	$\{-3, 4\}$	$\{-8, 6\}$	$\{-12, 9\}$	$\{-11, -4\}$	$\{-8, -1\}$

9/9. The same construction, re-pointed to a different level, recovers a different eigenform eigenvalue-for-eigenvalue.

6 The non-rational form: level 61

At $\mathfrak{p}_{61}$ ($N = 61$, split) the A_5 -action on $\mathbb{P}^1(\mathbb{F}_{61})$ has *three* orbits, $\mu = (12, 20, 30)$, $h = 3$, so the cuspidal dimension is 2. By [7, Rem. 4.4], 61 is the smallest level whose newform has coefficients outside $\mathbb{Q}$: the associated Shimura curve has genus 2, and its Jacobian is a modular abelian surface with real multiplication by $\mathbb{Q}(\sqrt{5})$. The $h = 2$ shortcut of §3 does not apply; instead we read the eigenvalue off the 3×3 matrix.

The all-ones vector is the Eisenstein eigenvector (eigenvalue $N(q) + 1$), so the characteristic polynomial of $B(q)$ factors over $\mathbb{Q}$ as $(x - (N(q) + 1))g_q(x)$, where $g_q(x) = x^2 - t_q x + n_q$ is the characteristic polynomial of the 2-dimensional cuspidal block. Comparing traces and determinants,

$$t_q = \text{tr } B(q) - (N(q) + 1), \quad n_q = \frac{\det B(q)}{N(q) + 1} \in \mathbb{Z}, \quad (2)$$

a priori two rational integers. Whether the two cuspidal eigenvalues are rational (two forms) or an irrational conjugate pair (one form with real multiplication) is then decided by the discriminant $t_q^2 - 4n_q$ — which we compute, rather than assume.

Verified (computational, reproducible)

Real multiplication, reconstructed. From (2) the discriminant $t_q^2 - 4n_q$ equals $5 \cdot (\text{square})$ at every prime $N(q) \in \{4, 9, 11, 19, 29, 49\}$. Hence g_q is irreducible over $\mathbb{Q}$ with roots $a_q, \sigma(a_q) \in \mathbb{Z}[\varphi]$ a Galois-conjugate pair: a *single* eigenform with real multiplication by K , not two rational forms. At the inert primes (one prime per rational prime, hence unambiguous) the eigenvalue matches [7] exactly:

$$a_4 = 2\varphi - 2, \quad a_9 = -\varphi - 2, \quad a_{49} = -4\varphi + 2,$$

i.e. $(t_q, n_q) = (-2, -4), (-5, 5), (0, -20)$ respectively. The Ramanujan bound holds on both real embeddings throughout.

The real multiplication is not merely consistent with the data; it follows from it:

Proposition 1. *If g_q has non-square discriminant for even a single q , then the 2-dimensional cuspidal Brandt module at level $\mathfrak{p}_{61}$ is irreducible over $\mathbb{Q}$, the (commutative) Hecke algebra acts on it through the quadratic field $\mathbb{Q}(\sqrt{t_q^2 - 4n_q})$, and every Hecke eigenvalue lies in that field. In*

particular, $t_4^2 - 4n_4 = 20 = 5 \cdot 2^2$ gives $\mathbb{Q}(\sqrt{5})$, so the level-61 eigenform has real multiplication by $\mathbb{Q}(\sqrt{5})$ and all its eigenvalues lie in $\mathbb{Z}[\varphi]$.

Proof. On the 2-dimensional cuspidal space, $t_q^2 - 4n_q$ non-square makes g_q irreducible over $\mathbb{Q}$, so T_q acts irreducibly. Any operator commuting with such an irreducible quadratic operator lies in $\mathbb{Q}[T_q] \cong \mathbb{Q}(\sqrt{t_q^2 - 4n_q})$, a field; the Hecke operators all commute and preserve the space, so they lie in $\mathbb{Q}[T_q]$, and every cuspidal Hecke eigenvalue lies in $E := \mathbb{Q}(\sqrt{t_q^2 - 4n_q})$. The value $t_4^2 - 4n_4 = 20$ fixes $E = \mathbb{Q}(\sqrt{5})$; integrality of the Brandt matrices puts the eigenvalues in $\mathcal{O}_K = \mathbb{Z}[\varphi]$. $\square$

So the geometry does not merely *count* the dimension-2 space at 61, nor merely match six published values: it *proves*, from a single discriminant, that the form has real multiplication by $\mathbb{Q}(\sqrt{5})$, and reconstructs its irrational Hecke eigenvalues.

7 The closure certificate algebra

The recurring structural fact above is that each Hecke operator carries a positive invariant form. We isolate this as a small certificate algebra, both because it is the cleanest statement of what the geometry guarantees and because it is the bridge to §8.

Definition 2 (Positive symmetrizer). *A positive symmetrizer for $T \in M_n(\mathbb{R})$ is a symmetric positive definite G with $GT = T^\top G$. Then T is self-adjoint for the inner product $\langle u, v \rangle_G = u^\top Gv$, so it has real spectrum and a G -orthonormal eigenbasis.*

(This is the self-adjoint, “lossless” end of a broader closure taxonomy — dissipative $T^\top GT \prec G$, isometric $T^\top GT = G$ — which we do not need here: the Brandt operators are not contractions, having Eisenstein eigenvalue $N(\mathfrak{q}) + 1 > 1$.)

Proposition 3. *For every $\mathfrak{q} \nmid \mathfrak{p}$ the Brandt operator $B(\mathfrak{q})$ admits the positive symmetrizer $G = \text{diag}(\mu)$, hence has real spectrum and a μ -orthonormal eigenbasis.*

Proof. The raw incidence count $C_{IJ} = \#\{(x, \gamma) : x \in \text{orbit } I, N(\gamma) = \varpi, x\iota_{\mathfrak{p}}(\gamma) \in \text{orbit } J\}$ satisfies $C_{IJ} = C_{JI}$ via the involution $(x, \gamma) \mapsto (x\gamma, \bar{\gamma})$ (using $\gamma\bar{\gamma} = N(\gamma)$ acting as a scalar on $\mathbb{P}^1$). Since $B(\mathfrak{q})_{IJ} = C_{IJ}/(120\mu_I)$ by (1), this gives $\mu_I B(\mathfrak{q})_{IJ} = \mu_J B(\mathfrak{q})_{JI}$, i.e. $GB(\mathfrak{q}) = B(\mathfrak{q})^\top G$. $\square$

Real spectrum is thus elementary. The sharper *Ramanujan bound* $|a_{\mathfrak{q}}| \leq 2\sqrt{N(\mathfrak{q})}$ that we observe at every tested prime is *not* elementary: it is the Ramanujan–Petersson bound for these Hilbert cusp forms, transported through Jacquet–Langlands and proved by Eichler–Shimura/Deligne [9, 8]; in this paper we record it as a property satisfied by the computed eigenvalues, not as something we prove. The Brandt operators are, in short, the *positive, finite, real-spectrum* fragment of the arithmetic.

8 Outlook: a heuristic relation to the Riemann Hypothesis

This section is explicitly heuristic; nothing in §3–§7 depends on it, and it contains no new theorem. We include it only to state precisely what the verified structure does *not* reach.

The rigorous statement. For an L -function, RH is equivalent to genuine positivity criteria: the Weil explicit-formula positivity [14] and Li’s criterion $\lambda_n \geq 0$ for all n [10, 3]. These are theorems (equivalences); they concern the *global, analytic* L -function and an *infinite* family of positivity conditions.

What the geometry delivers, and what it does not.

Verified (computational, reproducible)

The icosian construction delivers the *coefficient side*: the Hecke eigenvalues a_q , equivalently the local factors of the automorphic L -functions, with real spectrum (Prop. 3) and the observed Ramanujan bound. This is finite, per-prime, local data.

Open / not claimed

RH / GRH is the global, analytic statement that the assembled L -function has no off-line zero — the infinite positivity of Weil/Li. A finite Brandt computation produces eigenvalue *coefficients*, not zeros, and supplies none of this positivity. Programs that would close the gap — a self-adjoint operator with the zeros as spectrum (Hilbert–Pólya), or Connes’s arithmetic-site trace formula [4] — remain open; we make no contribution to them. Mirror symmetry (the functional equation) alone does not force zeros onto the line: the Davenport–Heilbronn phenomenon [6] exhibits Dirichlet series with the functional equation yet off-line zeros.

Interpretation (non-load-bearing)

Informally, one may picture the finite, self-adjoint Brandt operators as a “lossless resonator” and the conjectural absence of off-line zeros (the GUE statistics of [11, 1], itself a conjecture) as the infinite-dimensional analogue. This is a picture for orientation only. It is not an equivalence, not a theorem, and not used anywhere above.

9 Reproducibility

All results are produced by pure-Python scripts in the accompanying bundle (`route_b/`); none requires a computer-algebra modular package. The anti-circularity discipline is enforced in code: targets are computed by `point_count_target.py` (brute-force $\#E(\mathcal{O}_K/\mathfrak{q})$) and read only at comparison time.

script	result
<code>ideal_classes.py</code>	level-31 orbits, mass $8/15$, $h = 2$
<code>brandt_matrices.py</code>	Brandt/Hecke operators (1), invariants
<code>hecke_compare.py</code>	level 31, 24/24 out-of-sample ($N \leq 200$)
<code>second_form_level41.py</code>	level 41, 9/9 out-of-sample vs E_{41}
<code>genus2_form_level61.py</code>	level 61 genus-2, RM eigenvalues
<code>multilevel_dimensions.py</code>	cuspidal-dimension sequence (Table 1)
<code>no_fit_guard.py</code>	anti-circularity check

10 Conclusion

We have shown, by an independent geometric route, that the 600-cell / icosian order reproduces a *family* of Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$: a non-obvious dimension sequence (7/7), the full Hecke systems of two rational eigenforms (out-of-sample, no fit), and the irrational eigenvalues of the genus-2 form at level 61. We have framed the self-adjoint Hecke operators in a closure/resonance certificate algebra and used it to state exactly what separates this verified, finite, coefficient-side structure from the Riemann Hypothesis: the global analytic positivity, the lossless self-adjoint engine, which we characterize and do not construct. The value of the work is a clean, reproducible realization of known arithmetic from polytope geometry, and an honest account of the gap that remains.

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